The Aqua Cycle

Go Green, Go Fresh

Technical Documentation:

Model Structure: App Designs
We used proto.io to design the app. Proto.io is a cloud-based prototyping tool that allows users to design, build, and test interactive prototypes for mobile apps.
The screenshot of the App prototype is attached in the appendix.

The app prototype has an initial sign-in/sign-up landing page that would display the company's name and logo. It also includes a brief explanation about the app's function. When you chose to log in or sign up, it would redirect you to the log-in page or sign-up page. You can also access the My Profile page, which would display your information such as name, location, points, and number of refills. You can also navigate to an interactive map that shows the location of water fountains across the city. You can navigate to the company page and read about the company and our mission statement.

Developing the App

- **Development Approach**: We decided to use Swift/Objective-C for iOS or Java/Kotlin for Android as a hybrid app (e.g., React Native, Flutter) or as a web app.
- **Writing the Code**: We will collaborate with developers to ensure the design specifications are implemented correctly.
- **Integrate APIs and Backend**:
  - Connect to any backend services (e.g., databases, cloud storage, or APIs).
  - Implement functionalities such as user authentication, data management, or push notifications.
- **Optimize for Platforms**: We will ensure the app adheres to platform guidelines (e.g., Apple's Human Interface Guidelines or Google's Material Design).

Data Sources:

For our research and data collection, we have considered secondary sources and primary sources, and we intend to get more feedback about the usability of the app.

Secondary sources:

1. Water portability data

The dataset used in this research is crucial as it provides a comprehensive understanding of the factors influencing water potability, such as pH levels, hardness, chloramines, sulfate, turbidity, and organic carbon. These parameters are critical for assessing water safety and public health, enabling the identification of unsafe water sources and guiding effective treatment processes. The data drives the development of predictive models to evaluate potability, supporting data-driven decision-making for real-time monitoring and resource allocation.

This research aligns with the United Nations' SDG #6: Clean Water and Sanitation by offering actionable insights to improve water quality standards and reduce waterborne diseases. Ultimately, this dataset forms the foundation for meaningful contributions to public health, environmental sustainability, and scientific innovation.

2. Smart water fountain

The data generated by the Smart Water Fountain system is highly useful for research because it demonstrates how IoT can enable efficient water management through real-time monitoring and control. The **ultrasonic sensor data** provides insights into water levels, helping to optimize water usage and prevent wastage. By integrating this data with the **web interface**, researchers can explore the potential for remote interaction in resource management systems. Additionally, the simulation in Wokwi allows for iterative testing and refinement of IoT systems in a controlled environment, reducing the need for costly physical prototypes while ensuring the reliability and scalability of the data-driven system. This makes it a valuable tool for studying automation and sustainable resource management in IoT projects.

3. The data from the Massachusetts municipal solid waste and recycling program is critically important for research focused on environmental sustainability, waste management practices, and policy development. It provides comprehensive insights into the amounts and types of waste generated, recycled, and disposed of across cities and towns. This granular data allows researchers to:

1. **Analyze Trends**: Understand changes in waste generation and recycling rates over time.
2. **Evaluate Effectiveness**: Assess the impact of local and state-level waste management policies.
3. **Benchmark Practices**: Compare recycling and waste reduction efforts across municipalities to identify best practices.
4. **Model Sustainability Goals**: Use the data to design and test models for achieving zero-waste or improved recycling outcomes.
5. **Public Awareness**: Highlight areas requiring increased community engagement in waste reduction initiatives.

By providing actionable insights, this dataset supports efforts to enhance sustainability, reduce environmental impacts, and promote resource-efficient practices, making it a cornerstone for any research related to waste management and policy-making.

Preprocessing

The data for this research was imported into Google Colab, a cloud-based platform that provides a Python environment for running machine learning and data analysis tasks. By utilizing Python codes, various operations were performed on the dataset, including data preprocessing, visualization, and building predictive models.

Google Colab's powerful computational resources and support for libraries like Pandas, Matplotlib, and Scikit-learn enabled efficient exploration of the data and development of machine learning algorithms. This approach allowed for real-time code execution, collaborative research, and seamless integration of data and tools, making it an ideal platform for conducting this research.

AI Integration in the Aqua Cycle Project

The Aqua Cycle Project integrates AI to enhance the functionality and effectiveness of its MVP by leveraging IoT-enabled smart water fountains and an intelligent Fountain App. This combination of AI tools and data-driven insights ensures optimal performance and user engagement while addressing environmental sustainability.

Smart Water Fountains with IoT Integration

Description: IoT-enabled sensors are installed in water fountains to monitor real-time usage, water quality, and operational status.

AI Role:

- **Predictive Analytics with Machine Learning (ML)**: AI models analyze data from foot traffic, historical usage, and environmental conditions to identify optimal fountain locations and forecast water demand. This ensures better placement and resource allocation for maximum utility.
- **Time-Series Analysis for Maintenance**: Using time-series data, AI predicts maintenance needs, such as filter changes or mechanical repairs, before issues arise. This minimizes downtime and ensures consistent fountain functionality.

Fountain App Integration

Description: A mobile application provides users with real-time fountain locations, personalized sustainability dashboards, and gamified rewards for promoting eco-friendly behavior.

AI Role:

- **Geolocation and recommendations**: AI-powered recommendation systems analyze user behavior, location, and historical patterns to suggest nearby fountains and provide tailored usage tips.
- **Personalized Dashboards**: The app displays the user's sustainability impact (e.g., plastic bottles avoided), driven by AI insights from aggregated usage data, making users more aware of their contributions to environmental goals.
- **NLP-Powered Chatbots and Voice Navigation**: Natural Language Processing (NLP) enables conversational chatbots and voice-based navigation, allowing users to easily locate fountains, access tips, and resolve queries in a user-friendly manner.
- **Gamified Rewards**: Machine learning personalized rewards such as coupons or achievements, encouraging frequent usage by aligning with user preferences and behaviors.

Impact of AI on Solution Effectiveness

The integration of AI enhances the Aqua Cycle Project by:

- **Optimizing Resource Allocation**: ML models ensure efficient placement of fountains and reliable operation through predictive maintenance, reducing costs and increasing user satisfaction.
- **Enhancing User Engagement**: Personalized dashboards, recommendations, and gamified rewards drive user participation, making the app a key motivator for sustainable habits.
- **Minimizing Environmental Impact**: AI-powered insights track and promote sustainability efforts, emphasizing the project's mission to reduce plastic waste.
- By combining IoT data, ML models, and AI-driven app features, the Aqua Cycle Project provides a scalable, impactful solution to tackle plastic waste through innovative water fountain management.

Testing and Refinement Insights

We conducted testing by providing access to the prototype to selected people, and below is the potential feedback we received.

Outcomes of Initial Testing Rounds:

- **Key Challenges**:
  - **Sensor Calibration Issues**: IoT sensors in smart water fountains initially struggled with accuracy in detecting water quality and usage, leading to unreliable data.

    - **User Confusion with App Features**: Early testers reported difficulty navigating the app's sustainability dashboard and understanding gamified elements.
    - **Maintenance Delays**: Predictive maintenance models lacked sufficient data to provide accurate alerts, resulting in minor operational disruptions.
- **User Feedback**:
    - Users emphasized the need for **simpler app interfaces** and clearer instructions for accessing fountain information and rewards.
    - Many requested **real-time notifications** for nearby fountains when they were within walking distance.
    - A significant portion valued seeing the environmental impact in tangible metrics, such as the number of plastic bottles avoided.

Improvements Implemented:

- **Enhanced IoT Sensors**:
    - Sensors were recalibrated and fine-tuned to ensure accurate readings for water quality and usage patterns.
    - Additional testing ensured seamless integration of IoT data with AI models for real-time monitoring.
- **App Refinements**:
    - Simplified UI for the sustainability dashboard, making it more intuitive for users to track their impact and rewards.
    - Added **push notifications** and **voice navigation options** for locating nearby fountains.
- **Improved predictive models**:
    - Time-series data collection was extended during testing to refine the accuracy of predictive maintenance models.
    - The AI system now prioritizes high-traffic fountains for frequent checks, ensuring minimal downtime.

Key Insights Gained:

- **User Engagement requirements Simplicity**: A streamlined app interface and clear metrics significantly improved user satisfaction and engagement.
- **Data Sufficiency is Crucial for AI Models**: More extensive datasets enhanced the reliability of predictions for both fountain maintenance and usage optimization.
- **Real-Time Feedback Drives Usage**: Notifications and personalized tips increased user interaction, making the system more effective in promoting sustainability habits.

Contributions to Final MVP Iteration:

- A **fully calibrated IoT system** ensured reliable data for fountain operation and maintenance.
- The app featured **intuitive dashboards**, **gamified rewards**, and **real-time notifications**, aligning with user needs.
- Enhanced predictive models improved resource allocation and operational efficiency, ensuring a robust and scalable solution.

These refinements ensured that the Aqua Cycle Project's MVP effectively addresses user needs, operational challenges, and sustainability goals.

References

Neesha12. (n.d.). *Water Potability Analysis* [Kaggle Notebook]. Retrieved from https://www.kaggle.com/code/neesha12/water-potability-analysis/input

OpenAI. (2024). *ChatGPT (December 2024 version)* [Large language model]. OpenAI. https://openai.com/chatgpt

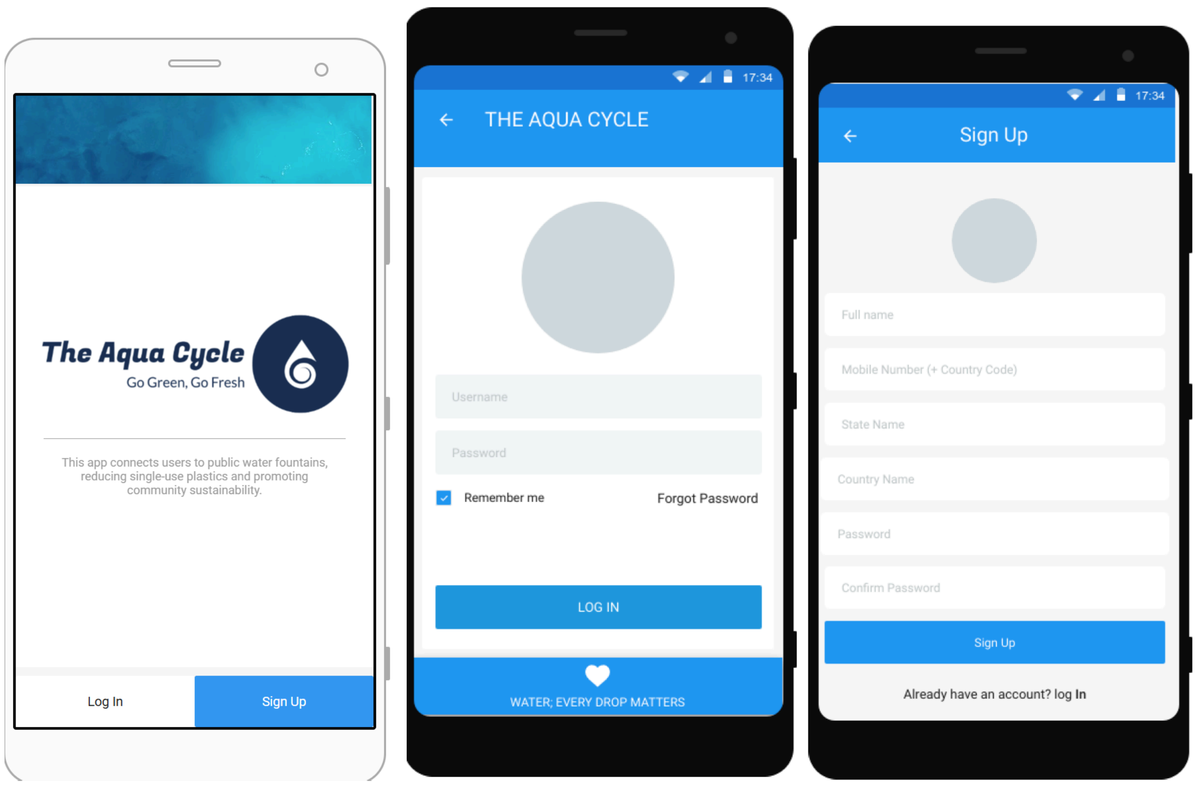
Proto.io. (n.d.). *Interactive prototype*. Retrieved December 14, 2024, from https://share.proto.io/1CNA4G/

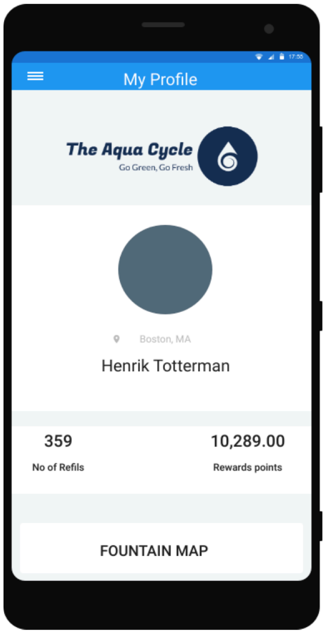
Ayishabanu, R. (n.d.). IOT Phase 3 [PDF]. GitHub. https://github.com/R-Ayishabanu/IOT/blob/main/IOT_Phase%203.pdf.

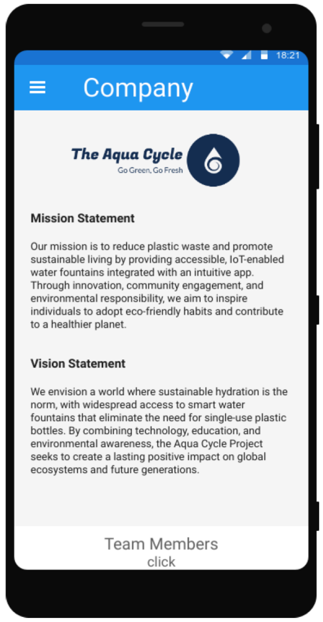
Massachusetts Department of Environmental Protection. (n.d.). *Recycling & solid waste data for Massachusetts cities & towns*. Retrieved December 15, 2024, from https://www.mass.gov/lists/recycling-solid-waste-data-for-massachusetts-cities-towns#municipal-solid-waste-&-recycling-data

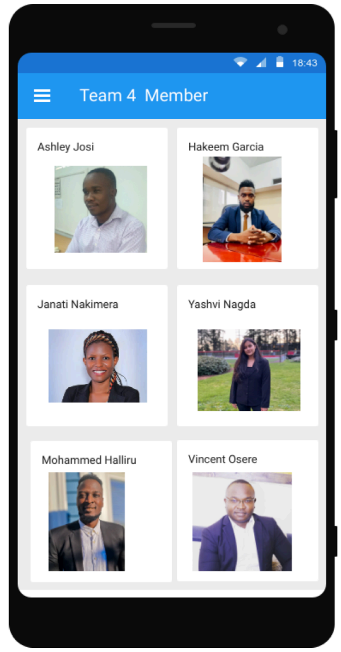
Appendices:

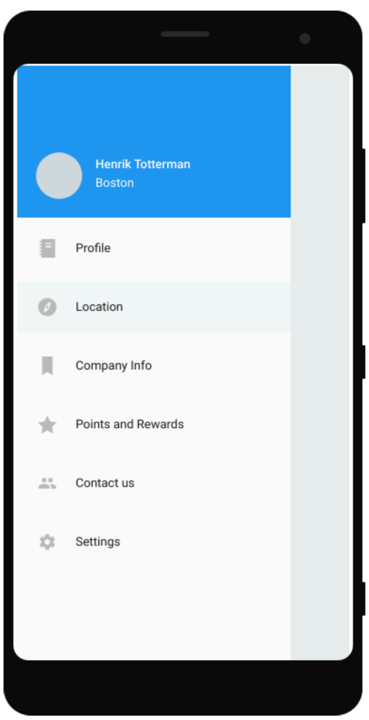
App prototype View: Link(https://pr.to/1CNA4G/)

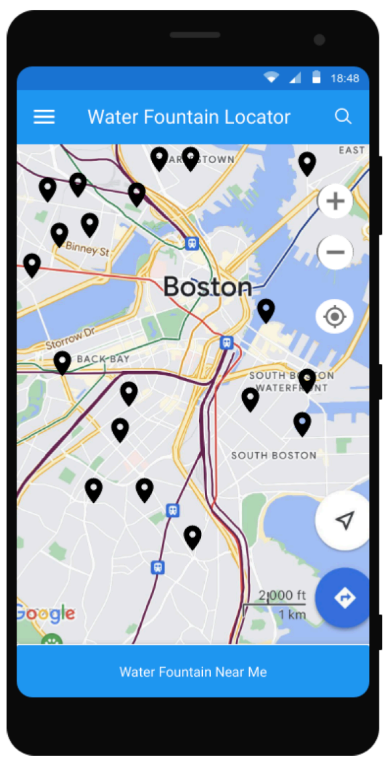


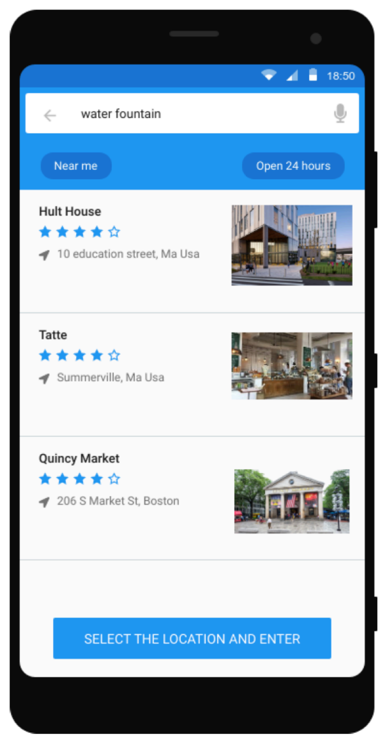

My Profile
The Aqua Cycle
Go Green, Go Fresh
Boston, MA
Henrik Totterman
359
No of Refils
10,289.00
Rewards points
FOUNTAIN MAP


18:21
Company
The Aqua Cycle
Go Green, Go Fresh
Mission Statement
Our mission is to reduce plastic waste and promote sustainable living by providing accessible, IoT-enabled water fountains integrated with an intuitive app. Through innovation, community engagement, and environmental responsibility, we aim to inspire individuals to adopt eco-friendly habits and contribute to a healthier planet.
Vision Statement
We envision a world where sustainable hydration is the norm, with widespread access to smart water fountains that eliminate the need for single-use plastic bottles. By combining technology, education, and environmental awareness, the Aqua Cycle Project seeks to create a lasting positive impact on global ecosystems and future generations.
Team Members
click

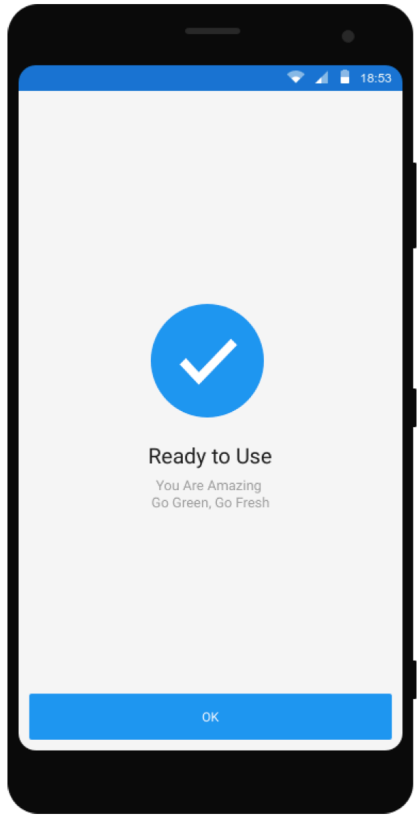

18:43
Team 4 Member
Ashley Josi
Hakeem Garcia
Janati Nakimera
Yashvi Nagda
Mohammed Halliru
Vincent Osere

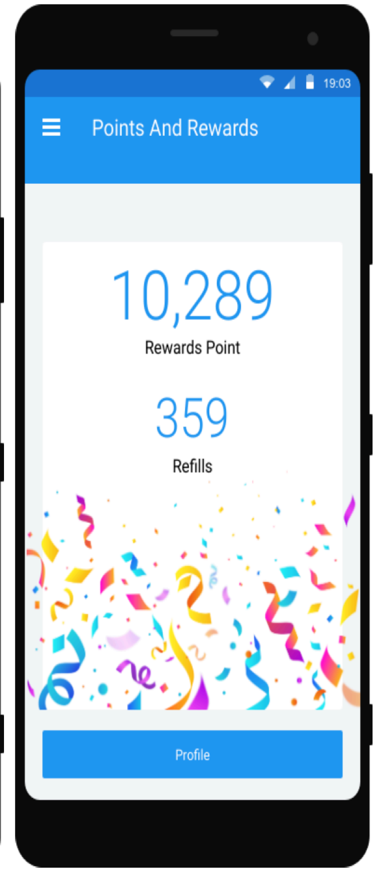

Henrik Totterman
Boston
Profile
Location
Company Info
Points and Rewards
Contact us
Settings

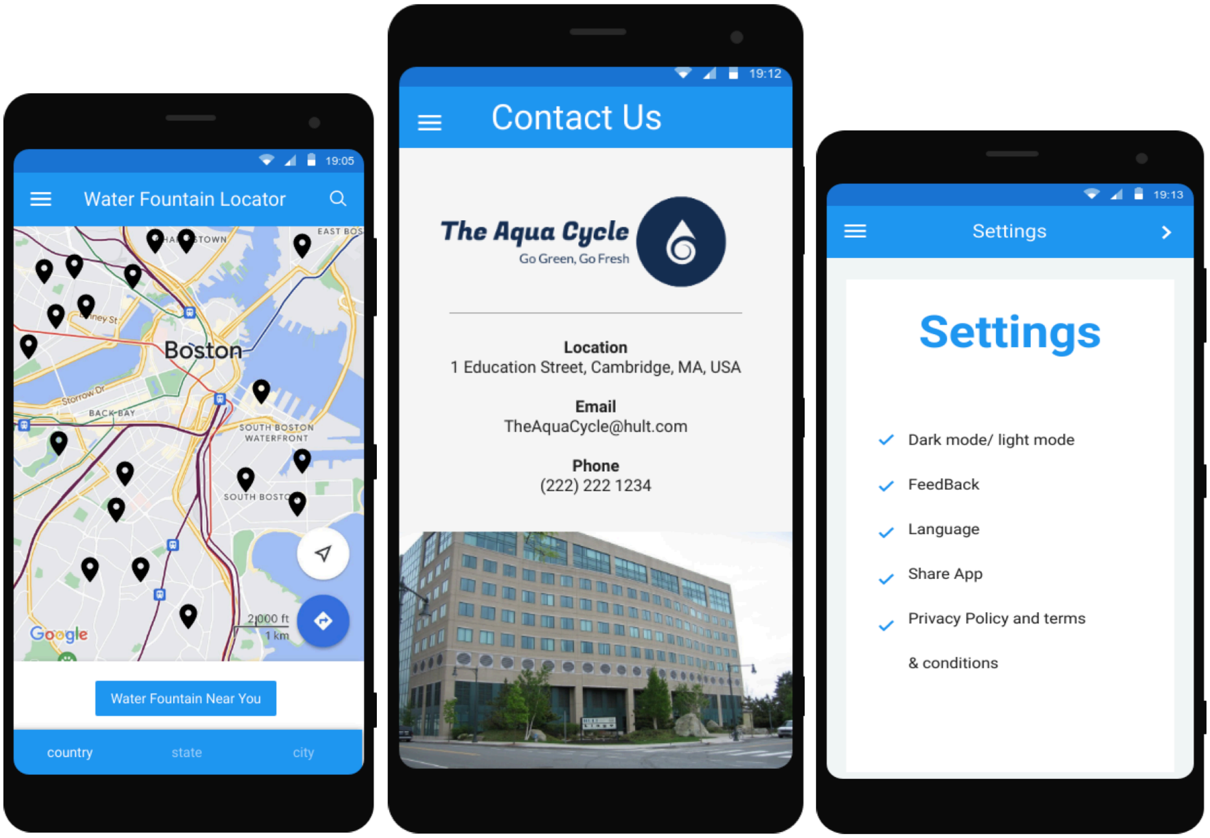

18:48
Water Fountain Locator
Boston
Binney St
Storrow Dr
BACK BAY
SOUTH BOSTON
SOUTH BOSTON WATERFRONT
EAST
2,000 ft
1 km
Google
Water Fountain Near Me


18:50
water fountain
Near me
Open 24 hours
Hult House
10 education street, Ma Usa
Tatte
Summerville, Ma Usa
Quincy Market
206 S Market St, Boston
SELECT THE LOCATION AND ENTER


Hult House, Cambridge
MA< USA
Align the QR Code within frame to scan
OK


18:53
Ready to Use
You Are Amazing
Go Green, Go Fresh
OK


19:03
Points And Rewards
10,289
Rewards Point
359
Refills
Profile


19:05
Water Fountain Locator
Boston
Google
2,000 ft
1 km
Water Fountain Near You
country
state
city
19:12
Contact Us
The Aqua Cycle
Go Green, Go Fresh
Location
1 Education Street, Cambridge, MA, USA
Email
TheAquaCycle@hult.com
Phone
(222) 222 1234
19:13
Settings
Settings
Dark mode/ light mode
FeedBack
Language
Share App
Privacy Policy and terms
& conditions